Approval-Based Elections in the 1D Euclidean Domain

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This dissertation is submitted for the degree of
Master of Science

March 2021

TODO

0.1 Resources

https://github.com/jasieksz/master-thesis-code

0.2 My Questions

1. ?

0.3 10/12/2020

1. Refactor code - vcr ilp detection, profile generation
2. Generate and analyze profiles: 5x5, 4x6, 6x4, random NxM

0.4 17/12/2020

1. Implement ilp MAV
2. Implement VR/CR MAV Liu and Guo (2016)
3. Compare ilp and poly MAV

0.5 07/01/2021

1. VR/CR MAV Liu and Guo (2016)

0.6 14/01/2021

1. VR/CR MAV Liu and Guo (2016)

0.7 11/02/2021

1. write thesis (+- 3 pages)
2. email about CR MAV to Liu/Guo
3. fix CR MAV algo
4. try to proof CR MAV
5. single thread experiments

0.8 18/02/2021

1. investigate CR MAV
2. VCR MAV ideas

3. email about CR MAV to Liu/Guo
4. try to proof CR MAV
5. write thesis
6. 20x20 VCR profiles - random distribution analysis
7. delete some random Candidates for VCR profile to become VR / CR

Abstract

(WIP)

I study approval-based elections, where preferences are in the 1D-Euclidean domain. I explore multiple areas, notably:

1. I create domain detection algorithms and evaluate the results.
2. I look for the efficient method of computing election results.

For the first issue I find ???. For the second one I create polynomial complexity algorithm (not yet) and compare it with LP solutions. (WIP)

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Chapter 1

Introduction

Elections are ubiquitous element of our society in one form or another. For example choosing members of parliament or a winner of a competition. In a day-to-day activities we need to decide on a meeting time or select recommended movies in a streaming service. Each of these choices may require different type of election and scoring rule, based on the underlying properties of the choice (i.e. voters should be represented proportionally by members of a government)

Reseach objectives

In this work I am going to focus on the *Approval Elections*, that is where each voter provides a subset of candidates that he approves of and disapproves the rest. In general many social choice problems are known to be computationally hard. However they become tractable under some restrictions on the structure of the preferences. Elkind and Lackner (2015) explored well known restrictions in the setting of approval preferences, introduced several new one-dimensional domain restrictions such as Dichotomous Euclidean (sprawdzic kiedy/kto wprowadził pojęcie - spatial pref, multidim unfolding) and derivatives (czy to dobre okreslenie na VR CR?). Faliszewski (2020) generalized this model for the N-dimensional euclidean space calling it Voter Candidate Range and explored two-dimensional case.

The goal of this work is to explore properties of the one-dimensional model. Search for the profile detection polynomial complexity algorithm. I will try as well to extend solution of Liu and Guo (2016) for computing MAV score with polynomial complexity to the VCR model.

Contributions

I provide linear programming preference detection algorithm and ...

Chapter 2

Preliminaries

In this chapter, I introduce the definitions and notations used in this work.

Approval Elections

Election is a pair $E = (C, V)$ where $C = \{c_1, ..., c_m\}$ is a set of candidates and $V = \{v_1, ..., v_n\}$ is a list of voters. Each voter $v \in V$ approves a subset of the candidates and disapproves the remaining candidates, I will be referring to it as a *approval set* and for each voter $v \in V$ denote it as $A(v)$, when using A just by itself I refer to all approval sets in the given election (preferences).

Committee Selection Rules

Voting rules are algorithms used to select winners (committee) of a given election. Each rule has different properties and must be selected accordingly to the goal of the election. For the detailed overview of multiwinner voting rules I point to Faliszewski et al. (2017).

Hamming Distance Minimax Approval Voting (MAV)

(WIP) dopisać wzory

Introduced by Brams et al. (2007). Selects a committee W that minimizes the maximum Hamming distance between W and voters' approval ballots. For an election E and a resulting committee of size k I denote winning committee as $W = MAV(E, k)$

Domain restrictions

(WIP) czy w ogóle o tym tutaj pisać??? czy wystarczy w Ch.3

Dichotmous Euclidean, Voter Interval, Candidate Interval

Elkind and Lackner (2015)

Single Peaked

Single Peaked On a Circle - Peters and Lackner (2020)

Single Crossing

Chapter 3

One Dimmensional Euclidean Domain

3.1 Euclidean Preference Restrictions

Voter Candidate Range

Given an election $E = (C, V)$, we say that the voters have *1D-Euclidean* preferences if for every agent (voter or candidate) there exists a point x_a and nonnegative real value radius r_a such that: For every candidate, voter pair: $c \in A(v) \Leftrightarrow \sqrt{(x_c - x_v)^2} \leq r_c + r_v$

We can easily think of it as: the point x_a represents ideal position in the 1-dimensional space of opinions. The radius can be seen as candidate's charisma or voter's willingness to compromise.

Candidate Range

We say that P satisfies CR if candidates in C can be ordered so that each of the sets v_i forms an interval of that ordering. CR is special case of VCR, where we require all the voters to have radii equal to zero

Voter Range

We say that P satisfies VR if the voters in P can be reordered so that for every candidate c the voters that approve c form an interval of that ordering. VR is a special case of VCR, where we require all the candidates to have radii equal to zero

Faliszewski (2020) showed that CR and VR are equivalent to respectively VI and CI introduced by Elkind and Lackner (2015).

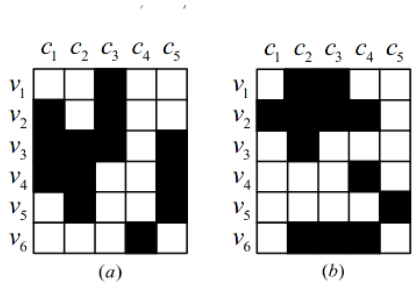

Fig. 3.1 CR and VR Temporary image

Larger Domain

The VCR model has larger domain than the VR and CR combined. For example, election with candidate set $C = \{a,b,c,d\}$ and voters with approval sets $\{a,b\}$, $\{b,c\}$, $\{b,d\}$, and $\{a,b,c,d\}$ is VCR, but neither VR nor CR.

3.2 Detection Algorithms (WIP)

Profile is VR or CR if the preference matrix or any permutation has consecutive ones property, that is for each row (VR) or column (CR), entries with 1 are in the uninterrupted sequence.

The fastest known algorithm for detection of C1P is based on the *PQ-tree* algorithms by Booth and Lueker (1976). However in this work I'm using still relatively fast and much easier to implement approach based on the *Lex-BFS* algorithm by Habib et al. (2000) (WIP) Change of the solution, using ILP instead - described in the ch.4

Voter Candidate Range detection algorithms are described in detail in the subsequent chapters.

3.3 (WIP) Committee Scoring Rules - Motivation

Explain how domain restrictions (VR/CR/VCR) can change complexity of computing rules, detection, etc. Swap section order with "Detecion Algorithms"

Section 4.1 in Faliszewski (2020)

"Elkind and Lackner (2015) have shown polynomial-time algorithms for computing CC winning committees in 1D-VR and 1D-CR elections. We unify this result into a single algorithm for 1D-VCR elections (the main idea is remarkably close to that for the 1D-VR case). Proposition 2. There is an algorithm that given a 1D-VCR election and committee size, computes some winning C committee in polynomial time. A natural question is whether it is possible to extend the above result to PAV and other Thiele methods. While we leave this question open, in Appendix B we explain that the existing technique that allowed to construct a polynomial-time algorithm for a large class of Thiele rules under the 1D-VR model (Peters and Lackner 2020) cannot be easily applied to the 1D-VCR setting. On the other hand, it may be possible to obtain a 1D-VCR algorithm for MAV as 1D-VR and 1D-CR algorithms do exist (Liu and Guo 2016)."

Chapter 4

Voter Candidate Range Profile Detection

This chapter describes in detail and compares created algorithms for the VCR property detection.

4.1 Overview

Detection algorithms are used to check whether election profile satisfies desirable structural properties. For example to check CR (VR) properties it's enough to check for the *Consecutive Ones Property* in the preferences matrix.

My focus lies on the one dimmensional VCR property. Having preferences matrix as input we have to provide a mapping of each agent (voter/candidate) to the point on real line and nonnegative number representing radius, so that it satisfies the VCR property.

4.2 Integer Linear Programming

ILP is a common approach to solve NP-hard problems, however it still can be used effictively for other problems. First we have to define the problem as an integer linear program. (WIP: Czy w ogole musze tlumaczyc co to ILP?) That is a system of linear inequalities and constraints and optionally linear function to optimize, alternatively we can just check feasibility of the constraints. To obtain a solution we have to use an optimization solver. Popular choices are Gurobi or CPLEX which are state of the art general purpose solvers.

Solution based on the ILP is excellent choice for the first algorithm to develop. It provides efficient and relatively simple detection algorithm. Additionally it can be used as a fast benchmark for checking future solutions.

Transform VCR definition into linear constraints

Before forumulating ILP we have to redefine the VCR definition beacasue it is based on the euclidean distance hence contains quadratic function. Equation 4.1 shows transforamtion of VCR equation into linear inequalities.

$$\begin{aligned} VCR &\iff \sqrt{(x_c - x_v)^2} \leq r_c + r_v \\ &\iff \|x_c - x_v\| \leq r_c + r_v \\ &\iff -(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \\ &\iff x_c - x_v \leq r_c + r_v \;\; \vee \;\; -(x_c - x_v) \leq r_c + r_v \end{aligned} \tag{4.1}$$

The constraint obtained from equation 4.1 is not suitable for ILP yet beacasue it contains logical *OR* condition which cannot be directly represented in the ILP. However this issue is quite common and most solvers provide additional mechanisms to solve it.

In Gurobi solver it's called *INDICATOR* constraint which allows to create versatile constraints like: $z = f \rightarrow a^T x \leq b$ which states that if the binary indicator variable z is equal to f, where $f \in \{0,1\}$, then the linear constraint $a^T x \leq b$ should hold. On the other hand, if $z = 1 - f$, the linear constraint may be violated.

Detection algorithm

Final formulation of the ILP for recognising VCR property in the eleciton profile is shown in figure 4.2. Variables used by the program represent agent's point on real line and their radius. Additionally I use two binary variables p and q required by the indicator constraints.

(WIP: Czy pisać jeszcze raz że approval set A jest jedyną daną wejściową?)

(WIP: Poprawić format zapisu ILP na bardziej standardowy)

$$
\begin{aligned}
&\textbf{check feasibility}: \\
&\quad \textbf{subject to}: \\
&\qquad x_a \in \mathbb{R} && \forall\, \mathrm{a} \in C \cup V \\
&\qquad r_a \in \mathbb{R}^+ && \forall\, \mathrm{a} \in C \cup V \\
&\qquad p_{cv} \in \{0,1\} && \forall\, \mathrm{c} \in \mathrm{C}\ \forall\, \mathrm{v} \in V \\
&\qquad q_{cv} \in \{0,1\} && \forall\, \mathrm{c} \in \mathrm{C}\ \forall\, \mathrm{v} \in V \\
&\quad \text{if } c \in A(v): \\
&\qquad x_c - x_v \leq r_c + r_v \\
&\qquad -(x_c - x_v) \leq r_c + r_v \\
&\quad \text{if } c \notin A(v): \\
&\qquad p_{cv} == 1 \rightarrow x_c - x_v \geq r_c + r_v + 1 \\
&\qquad q_{cv} == 1 \rightarrow -(x_c - x_v) \geq r_c + r_v + 1 \\
&\qquad p_{cv} + q_{cv} >= 1
\end{aligned}
\tag{4.2}
$$

4.3 (WIP) My VCR detection algorithm

4.4 Performance Comparison

How fast Gurobi Solver can be?

Chapter 5

Voter Candidate Range Profiles

In this section I explore VCR profiles domain and describe found properties. The main goal is to compare size of the domain with other well known domains such as VR, CR and others (WIP: Czy jeszcze jakies inne, SC, SP?).

5.1 Domain analysis - experiment

To achive the goal I've performed several experiments. For the first one I check how many profiles have VCR property but not VR nor CR. As the input I use every combination of the profiles for the given election size. For the second experiment I generate random VCR profiles and again check how many of them are not VR nor CR.

In both experiments I use detection algorithm described in the section 4.2. Additionally the second experiment is split into sub-experiments based on the different random distributions (WIP: Czy "distributions" jest ok?) used to generate profiles.

5.2 How many VCR profiles are not VR nor CR?

Experiment 1 - small profiles

For the small profiles I performed complete search of the domain. I checked the elections with the following sizes (M candidates N voters): $[(3,3),(4,4),(5,5),(4,6),(6,4)]$. The results as seen in the table 5.1 show that VCR domain is relatively large, however only limited number of profiles are not VR nor CR.

WIP : Prettify the table

WIP : Make some charts, write statistics, show examples

	3C 3V	4C 4V	5C 5V	4C 6V	6C 4V
Count	2^9	2^{16}	2^{25}	2^{24}	2^{24}
VCR	506	57832	19846082	10893736	10893736
VR	-	-	-	-	-
CR	-	-	-	-	-
VCR !COP	0	96	501000	144480	144480

Table 5.1 Small profiles NCOP/VCR

Profile examples

Example of 5C 5V election that is VCR but not VR nor CR.

```
Approval Set
    [1., 1., 0., 0., 1.]
    [1., 1., 1., 1., 1.]
    [1., 0., 0., 0., 0.]
    [0., 0., 0., 1., 1.]
    [0., 0., 1., 0., 1.]

Candidates
    Candidate(id='A', x=3.0, r=1.0),
    Candidate(id='B', x=2.0, r=0.0),
    Candidate(id='C', x=0.0, r=0.0),
    Candidate(id='D', x=1.0, r=0.0),
    Candidate(id='E', x=1.0, r=1.0)

Voters
    Voter(id='V0', x=2.0, r=0.0),
    Voter(id='V1', x=0.0, r=3.0),
    Voter(id='V2', x=3.0, r=0.0),
    Voter(id='V3', x=1.0, r=0.0),
    Voter(id='V4', x=0.0, r=0.0)
```

Experiment 2 - large profiles

For the larger size of profiles I generated a suitable number of random VCR profiles and checked how many of them are not VR nor CR. To create random VCR profile I generated a

point and radius for every voter and candidate and then created approval sets based on the VCR definition.

	10C 10V	15C 15V	20C 20V
VCR	1000	1000	3000
VR	-	-	-
CR	-	-	-
VCR !COP	24	667	2928

Table 5.2 Large profiles NCOP/VCR

5.3 (WIP) Experiment conclusions

Maybe do some nice graphs here and explore properties

Chapter 6

Minimax Scoring Rule Algorithms

6.1 Overview

Minimax Approval Voting (MAV) As close as possible to the preferences of all parties. By "close" we mean that the maximum distance of the compromise from the position of any state is minimal, which we call a minimax outcome. Brams et al. (2007) Selects a committee W that minimizes the maximum Hamming distance between W and voters' approval ballots. We define MAV-score of a committee W as $Max(f(W,v1),...,f(W,vn))$ where $f(W,vi) =$.

(WIP) Rewrite MAV definition, use easier equation formulation

MAV returns a committee W with the lowest MAV-score.

Brute force algorithm

```
def hammingDistance(a: np.array, b: np.array) -> int:
return (a != b).nonzero()[0].shape[0]

def mavScore(profile: Profile, committee:np.array) -> int:
    partialHD = lambda preference: hammingDistance(committee, preference)
    return max([partialHD(pref) for pref in profile.A])

def minimax(profile: Profile, k: int) -> np.array:
    candidateCount = len(profile.C)
    scores = [(mavScore(profile, np.array(committee)), profile)
                for committee in combinations(range(candidateCount), k)]
    scores.sort(key=lambda mav_p: mav_p[0])
    return scores
```

6.2 (WIP) MAV ILP from Lackner and Skowron (2020)

6.3 (WIP) Proof / Disproof? of MAV for VR and CR from Liu and Guo (2016)

6.4 (WIP) My algorithm for VCR profiles

(WIP) Current approches

Greedy

1. Sort by ?
2. ?

Dynamic

6.5 Comparison

Chapter 7

(WIP) Approval deletion for VCR profile to become VR or CR

7.1 Section One

7.2 Section Two

Chapter 8

Evaluation

8.1 Conclusions

8.2 Future Work

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